



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001:2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Electrical and Electronics Engineering Academic Year 2024-2025 (Odd Semester)

Degree, Semester & Branch : V Semester B.E EEE
Course Code & Title : CEI331 – PLC Programming
Name of the Faculty member : Mr. E. Thangam, AP(SG) - EEE

Innovative Practice Description

- **Unit / Topic:** Unit II / PLC to PC Communication and PC to PLC Communication
- **Course Outcome:** CO 2
- **Topic Learning Outcome:** TLO 8
- **Activity Chosen:** Demonstration
- **Justification:**

Enhanced Practical Understanding:

Demonstration provides a hands-on approach to learning, allowing students to see real-time data exchange between the PLC and PC. This bridges the gap between theoretical knowledge and practical application, making abstract concepts more tangible and easier to understand.

Engagement and Interactivity:

Watching the process in action, such as programming a PLC, observing data logs, or troubleshooting communication issues, keeps students engaged. It encourages active learning, where students can ask questions and participate in the setup, fostering a deeper grasp of the subject.

Skill Development for Real-World Application:

Demonstrations simulate real-world scenarios, preparing students for industry standards. By observing and interacting with the equipment, they gain confidence and competence in configuring, monitoring, and troubleshooting PLC-PC communication systems, which are critical skills in automation and control engineering.

- **Time Allotted for the Activity:** 30 minutes

- **Details of the Implementation:**

Hardware Setup:

Connect the PLC and PC using appropriate communication protocols (e.g., Ethernet). Ensure both devices are powered and configured for data exchange.

Software Configuration:

Install and configure software tools dedicated PLC programming environments (Siemens Logo Soft Comfort V8.3 on the PC. Set up communication parameters such as baud rate, IP address, and port settings.

Demonstration Activity:

Showcase data exchange by sending commands from the PC to the PLC and visualizing sensor or process data on the PC. Highlight troubleshooting steps for common communication issues.

- CO – PO / PSO mapping:**

CO	PO 1	PO 2	PO 3	PO 9	PO10	PO11	PO12	PSO 1	PSO 2	PSO 3
CO 2	3	2	3	1	1	1	1	1	2	3

(1 – Low 2 – Moderate 3 – High)

- PO / PSO mapped:**

Innovative practice	PO 9	PO 10	PO 12
	1	1	1
Justification for correlation	Demonstrating PLC-PC communication fosters teamwork as students collaborate to troubleshoot and configure systems, simulating real-world multidisciplinary tasks.	The activity enhances students' ability to document processes and convey technical details effectively during presentations and discussions.	The demonstration activity encourages students to adapt to evolving automation technologies, fostering a mindset of continuous learning to stay updated with industry advancements.

- Reflective Critique:**

- ❖ **Feedback of Practice from Students and Other Stakeholders:**

Students appreciated the hands-on learning approach, which clarified complex concepts. Stakeholders highlighted its relevance to industry practices. Suggestions included increasing the complexity of scenarios for advanced learning.

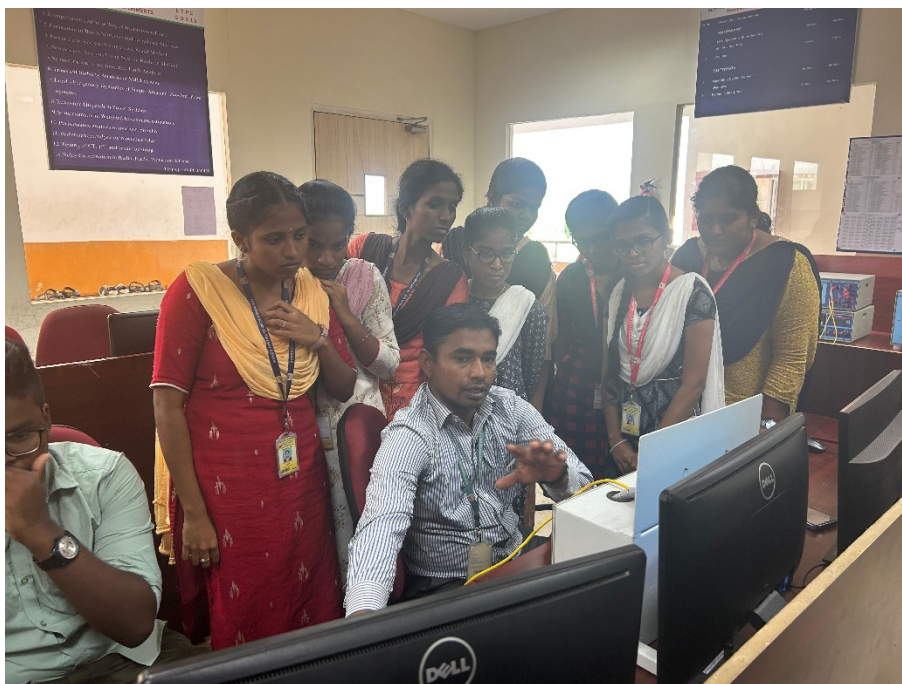
- ❖ **Benefit of the Practice:**

The activity bridged the gap between theoretical knowledge and practical skills. It enhanced students' confidence in handling real-world automation tasks and improved their employability in the automation sector.

- ❖ **Challenges Faced in Implementation:**

Ensuring compatibility between hardware and software was time-consuming. Limited availability of equipment restricted simultaneous practice. Some students faced initial difficulty in understanding communication protocols.

Photographic Evidence:



References:

1. Frank D. Petruzella, "Programmable Logic Controllers", 5th Edition, McGraw- Hill, New York, 2019.
2. Stuart Boyer A, "SCADA: Supervisory control and data Acquisition", Fourth Edition, ISA- The Instrumentation, Systems, and Automation Society, 2010

Signature of Faculty Member

HoD/EEE



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001:2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Electrical and Electronics Engineering Academic Year 2024-2025 (Odd Semester)

Degree, Semester & Branch : V Semester B.E EEE
Course Code & Title : CEI331 – PLC Programming
Name of the Faculty member : Mr. E. Thangam, AP(SG) - EEE

Innovative Practice Description

- **Unit / Topic:** Unit V / Exercise in Programming Languages from IEC 61131-3
- **Course Outcome:** CO 5
- **Topic Learning Outcome:** TLO 19
- **Activity Chosen:** Demonstration
- **Justification:**

Enhanced Practical Understanding:

Demonstration provides a hands-on approach to learning, allowing students to do Exercise in Programming Languages from IEC 61131-3. This bridges the gap between theoretical knowledge and practical application, making abstract concepts more tangible and easier to understand.

Engagement and Interactivity:

Watching the process in action, such as programming a PLC, observing data logs, or troubleshooting communication issues, keeps students engaged. It encourages active learning, where students can ask questions and participate in the setup, fostering a deeper grasp of the subject.

Skill Development for Real-World Application:

Demonstrations simulate real-world scenarios, preparing students for industry standards. By observing and interacting with the equipment, they gain confidence and competence in configuring, monitoring, and troubleshooting exercise in Programming Languages from IEC 61131-3, which are critical skills in automation and control engineering.

- **Time Allotted for the Activity:** 30 minutes

- **Details of the Implementation:**

Hardware Setup:

Connect the PLC and PC using appropriate communication protocols (e.g., Ethernet). Ensure both devices are powered and configured for data exchange.

Software Configuration:

Install and configure software tools dedicated PLC programming environments (Siemens Logo Soft Comfort V8.3 on the PC. Set up communication parameters such as baud rate, IP address, and port settings.

Demonstration Activity:

Showcase data exchange by sending commands from the PC to the PLC and visualizing sensor or process data on the PC. Highlight troubleshooting steps for common communication issues.

- CO – PO / PSO mapping:**

CO	PO 1	PO 2	PO 3	PO 9	PO10	PO11	PO12	PSO 1	PSO 2	PSO 3
CO 5	3	2	3	1	1	1	1	1	2	3

(1 – Low 2 – Moderate 3 – High)

- PO / PSO mapped:**

Innovative practice	PO 9	PO 10	PO 12
	1	1	1
Justification for correlation	Demonstrating PLC-PC communication fosters teamwork as students collaborate to troubleshoot and configure systems, simulating real-world multidisciplinary tasks.	The activity enhances students' ability to document processes and convey technical details effectively during presentations and discussions.	The demonstration activity encourages students to adapt to evolving automation technologies, fostering a mindset of continuous learning to stay updated with industry advancements.

- Reflective Critique:**

- ❖ **Feedback of Practice from Students and Other Stakeholders:**

Students appreciated the hands-on learning approach, which clarified complex concepts. Stakeholders highlighted its relevance to industry practices. Suggestions included increasing the complexity of scenarios for advanced learning.

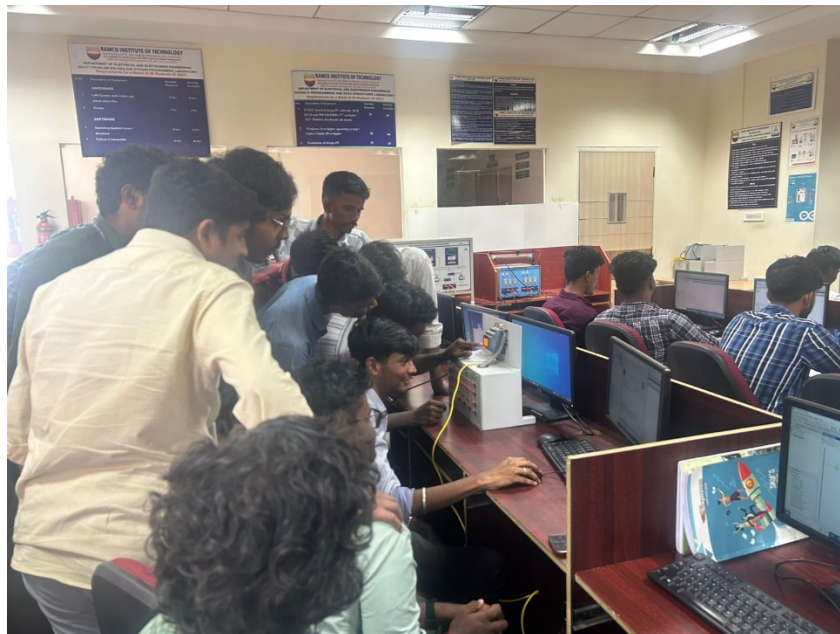
- ❖ **Benefit of the Practice:**

The activity bridged the gap between theoretical knowledge and practical skills. It enhanced students' confidence in handling real-world automation tasks and improved their employability in the automation sector.

- ❖ **Challenges Faced in Implementation:**

Ensuring compatibility between hardware and software was time-consuming. Limited availability of equipment restricted simultaneous practice. Some students faced initial difficulty in understanding communication protocols.

Photographic Evidence:



References:

1. Frank D. Petruzella, "Programmable Logic Controllers", 5th Edition, McGraw- Hill, New York, 2019.
2. Stuart Boyer A, "SCADA: Supervisory control and data Acquisition", Fourth Edition, ISA- The Instrumentation, Systems, and Automation Society, 2010

Signature of Faculty Member

HoD/EEE



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001:2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Electrical and Electronics Engineering Academic Year 2024-2025 (Odd Semester)

Degree, Semester & Branch : V Semester B.E EEE
Course Code & Title : CEI331 – PLC Programming
Name of the Faculty member : Mr. E. Thangam, AP(SG) - EEE

Innovative Practice Description

- **Unit / Topic:** Revision of Unit I to V
- **Course Outcome:** CO1 to CO 5
- **Topic Learning Outcome:** TLO 1 to TLO 21
- **Activity Chosen:** Mind Map
- **Justification:**

❖ Visual Organization of Concepts:

Mind maps help students organize and interrelate complex topics, such as PLC programming, communication protocols, and SCADA systems, providing a clear overview of their connections and applications.

❖ Active Learning and Creativity:

Creating a mind map fosters active engagement, encouraging students to think critically about the topics and creatively structure their knowledge.

❖ Facilitates Retention and Recall:

The visual representation aids memory retention, enabling students to recall intricate details about standards like IEC61131-3 and systems like SCADA and HART during assessments and real-world scenarios.

- **Time Allotted for the Activity:** 30 minutes

• Details of the Implementation:

❖ Introduction and Guidelines:

Explain the topics (PLC, IEC61131-3, HART, SCADA) and provide guidelines for creating a mind map. Highlight the importance of linking related concepts visually.

❖ Mind Map Creation:

Students individually or in groups create a mind map using tools like paper, whiteboards, or digital software. They organize concepts hierarchically with keywords, branches, and connections.

❖ Presentation and Discussion:

Students present their mind maps, explaining the interconnections. The teacher

provides feedback, clarifies misconceptions, and discusses real-world applications of the topics.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1	1	-	-	-	-	-	1	1	1	1
CO2	3	2	3	-	-	-	-	-	1	1	1	1
CO3	3	1	1	-	-	-	-	-	-	1	1	1
CO4	3	2	1	-	-	-	-	-	-	1	1	1
CO5	3	2	2	-	2	-	-	-	3	1	3	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO 9	PO 10	PO 12
	1	1	1
Justification for correlation	Demonstrating PLC-PC communication fosters teamwork as students collaborate to troubleshoot and configure systems, simulating real-world multidisciplinary tasks.	The activity enhances students' ability to document processes and convey technical details effectively during presentations and discussions.	The demonstration activity encourages students to adapt to evolving automation technologies, fostering a mindset of continuous learning to stay updated with industry advancements.

- **Reflective Critique:**

- **Feedback of Practice from Students and Other Stakeholders:**

Students found the mind map activity engaging and helpful for understanding connections between topics. Stakeholders appreciated its emphasis on concept clarity and visual learning. Suggestions included using advanced software for better presentations.

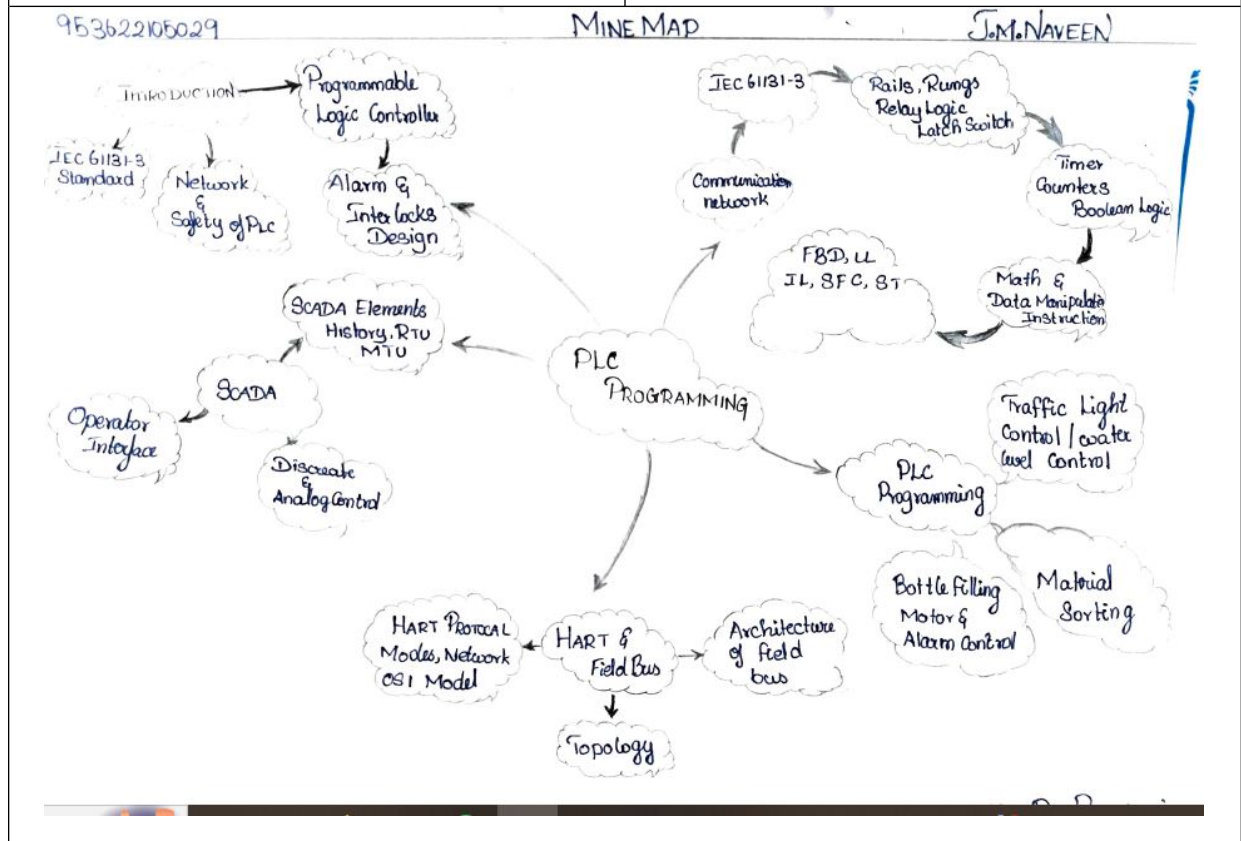
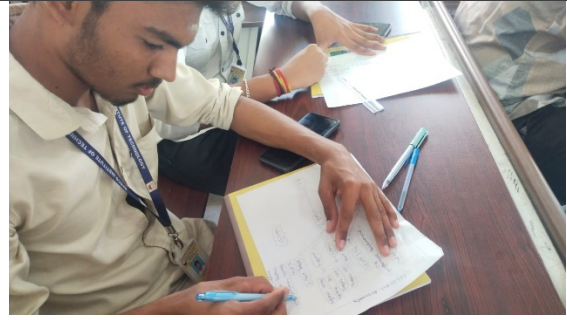
- **Benefit of the Practice:**

The activity enhanced critical thinking and facilitated a comprehensive understanding of complex topics. It encouraged collaborative learning and improved students' ability to organize and present technical information effectively.

- **Challenges Faced in Implementation:**

Some students struggled with structuring their mind maps effectively. Time constraints limited detailed exploration of each topic. Access to digital tools was uneven among participants.

Photographic Evidence:



References:

1. Frank D. Petruzella, "Programmable Logic Controllers", 5th Edition, McGraw- Hill, New York, 2019.
2. Stuart Boyer A, "SCADA: Supervisory control and data Acquisition", Fourth Edition, ISA- The Instrumentation, Systems, and Automation Society, 2010

Signature of Faculty Member

HoD/EEE



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001:2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Electrical and Electronics Engineering Academic Year 2024 - 2025 (Odd Semester)

Degree, Semester & Branch : V Semester B.E EEE
Course Code & Title : CEI331 – PLC Programming
Name of the Faculty member : Mr. E. Thangam, AP(SG) - EEE

Innovative Practice Description

- **Unit / Topic:** Unit III / History of SCADA
- **Course Outcome:** CO 3
- **Topic Learning Outcome:** TLO
- **Activity Chosen:** One Minute Paper
- **Justification:**

The One Minute Paper activity is ideal for teaching the history of SCADA as it encourages students to summarize key points concisely, reinforcing their understanding of its evolution and impact. This method promotes active learning and helps identify gaps in their knowledge. It's a quick, effective way to engage students and assess their grasp of the topic.

Photographer & Reporter: Myself

- **Time Allotted for the Activity:** 30 minutes
- **Details of the Implementation:**

- **Preparation and Setup**

To implement the One Minute Paper activity, the topic "History of SCADA" is introduced during a lecture or presentation. Key points about the evolution of SCADA systems, their significance in industrial automation, and technological milestones are covered in the session. Students are informed about the activity in advance, ensuring they pay close attention to the lecture. At the end of the session, students are provided with a prompt or question, such as "What are the key developments in the history of SCADA?" or "Why is SCADA important in industrial processes?"

- **Execution of the Activity**

After the lecture, students are given exactly one minute to write their responses to the prompt. They summarize their understanding in a few sentences, focusing on the main ideas and critical takeaways. The activity is conducted individually to ensure independent thought and reflection. To keep the task concise and focused, students are encouraged to highlight only the most significant points they recall, promoting active engagement and quick synthesis of information.

- **Review and Feedback**

Once the responses are collected, the instructor reviews them to gauge the students' understanding of the topic. Common themes and insightful answers are shared with the class, fostering a collaborative discussion. If any misconceptions or knowledge gaps are identified, they are addressed immediately. This step not only reinforces the content but also helps in adapting future lessons to better suit the students' learning needs.

- **CO – PO / PSO mapping:**

CO	PO 1	PO 2	PO 3	PO10	PO11	PO12	PSO 1	PSO 2	PSO 3
CO 3	3	1	1	1	1	1	1	2	3

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO 10	PO 11	PO 12
	1	1	1
Justification for correlation	Encourages understanding of modern engineering tools like SCADA for sustainable development.	Promotes ethical responsibility by highlighting SCADA's role in safe and reliable automation.	Fosters life-long learning by showcasing the evolving nature of SCADA technologies.

- **Reflective Critique:**

- ❖ **Feedback of Practice from Students and Other Stakeholders**

Students appreciated the concise and engaging nature of the activity, which helped them focus on key concepts. Teachers found it effective for assessing understanding and sparking interest in SCADA's history. Stakeholders valued the activity for promoting active learning and fostering better retention.

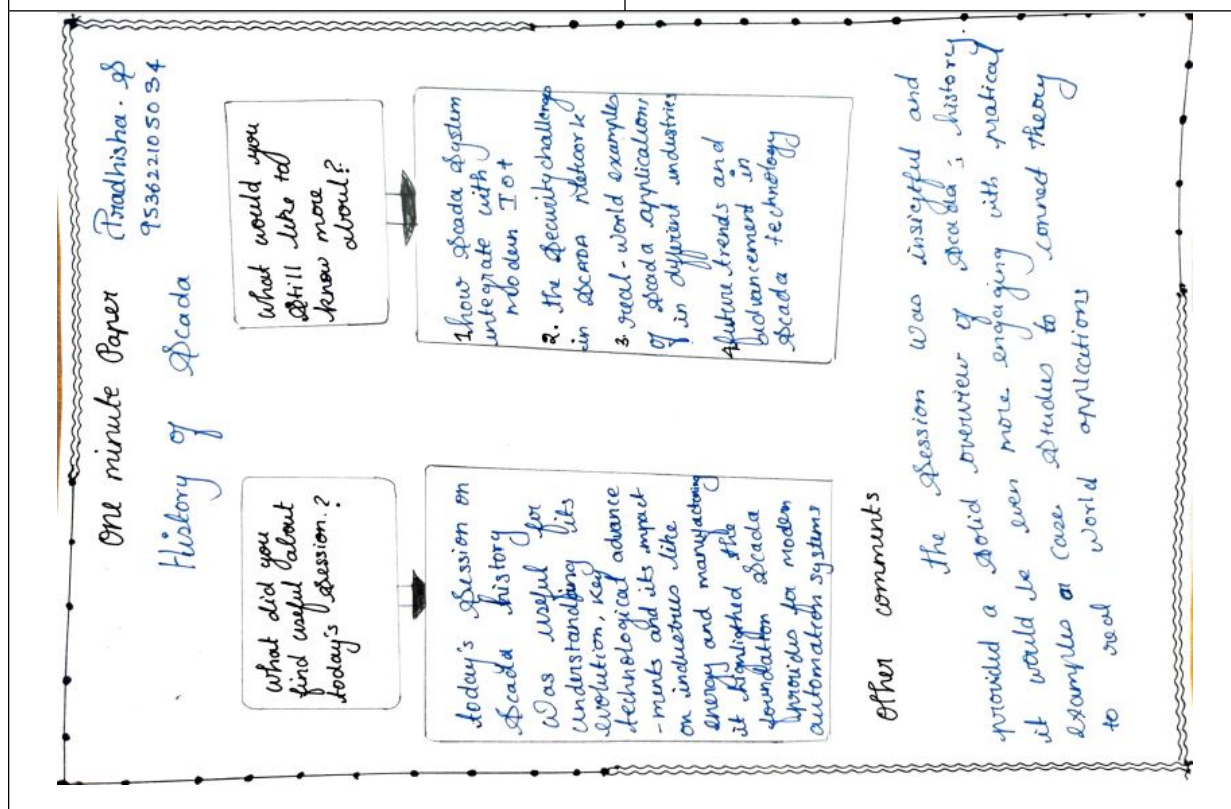
- ❖ **Benefit of the Practice**

The activity enhances critical thinking, encourages active participation, and provides immediate insights into students' comprehension. It helps students consolidate learning effectively while offering teachers a quick assessment tool. Additionally, it fosters a dynamic and interactive classroom environment.

- ❖ **Challenges Faced in Implementation**

Some students found it challenging to summarize complex information within the given time. Ensuring all students participated actively required additional encouragement. Limited time for detailed feedback during class posed a constraint for addressing all responses.

- **Photographic Evidence:**



References:

1. Frank D. Petruzella, "Programmable Logic Controllers", 5th Edition, McGraw- Hill, New York, 2019.
2. Stuart Boyer A, "SCADA: Supervisory control and data Acquisition", Fourth Edition, ISA- The Instrumentation, Systems, and Automation Society, 2010

Signature of Faculty Member

HoD/EEE



Department of Electrical and Electronics Engineering

Academic Year 2024 – 2025 (Odd Semester)

Degree, Semester & Branch: V Semester B.E. EEE

Course Code & Title: CEI331 PLC Programming

Name of the Faculty member (s): Mr. E. Thangam

Innovative Practice Description

- **Unit / Topic:** Unit I / Industrial Automation using PLC versus Microcontroller
- **Course Outcome:** CO 1
- **Topic Learning Outcome:** TLO – 1
- **Activity Chosen:** Think-Pair-Share
- **Justification:**

The Think-Pair-Share activity is justified for the topic "Industrial Automation using PLC versus Microcontroller" as it enables students to analyze the differences collaboratively, share diverse perspectives, and refine their understanding of selecting the right technology for specific industrial automation applications.

- **Time Allotted for the Activity:** 7 Minutes (Think – 2 Min, Pair – 3 Min, Share – 2 Min)
- **Details of the Implementation:**

Preparation and Introduction

The instructor begins by providing an overview of the topic, explaining the roles of PLCs and microcontrollers in industrial automation. Key points such as their functionalities, advantages, limitations, and typical applications are highlighted. Students are then introduced to the Think-Pair-Share activity, with clear instructions on the process and expected outcomes. The question for discussion might be: "Which is more suitable for industrial automation: PLC or microcontroller, and why?"

Execution of the Activity

1. **Think:** Each student is given 2-3 minutes to independently reflect on the question and jot down their thoughts, focusing on key differences and real-world implications.
2. **Pair:** Students pair up with a partner to share their ideas and discuss their perspectives for about 5 minutes. This stage encourages mutual learning and critical evaluation of each other's points.

3. **Share:** Pairs then present their combined insights to the entire class. The instructor facilitates this stage by summarizing key takeaways and addressing any misconceptions.

Follow-up and Feedback

After the sharing stage, the instructor consolidates the discussion, emphasizing important concepts and providing additional context if needed. A short quiz or reflective question can be assigned to reinforce learning. Feedback is also collected from students to gauge their engagement and understanding, helping to refine the activity for future sessions.

Total Strength is 63,

Photographer: one student - Mr. Sriratchan (interested in photography)

Reporter: Myself

• CO – PO / PSO mapping:

CO	PO 1	PO 2	PO 3	PO9	PO10	PO11	PO12	PSO1	PSO 2	PSO 3
CO 1	3	1	1	1	1	1	1	1	2	3

(1 – Low 2 – Moderate 3 – High)

• PO / PSO mapped:

Innovative practice	PO9	PO 10	PO 11	PO 12
	1	1	1	1
Justification for correlation	Enhances the ability to work effectively in teams by promoting collaboration and peer discussion on complex topics.	Provides exposure to modern engineering tools like PLCs and microcontrollers, helping students apply theory to practical automation scenarios.	Develops ethical responsibility by encouraging thoughtful analysis of the advantages and limitations of automation technologies in real-world settings.	Encourages life-long learning by fostering critical thinking and adaptability in choosing between PLC and microcontroller-based solutions for industrial automation.

• Reflective Critique:

Feedback of Practice from Students and Other Stakeholders

Students appreciated the collaborative aspect, which allowed them to hear different viewpoints and refine their own understanding. Teachers observed that the activity

promoted critical thinking and active participation. Stakeholders felt the exercise helped bridge theoretical knowledge with practical application in industrial automation.

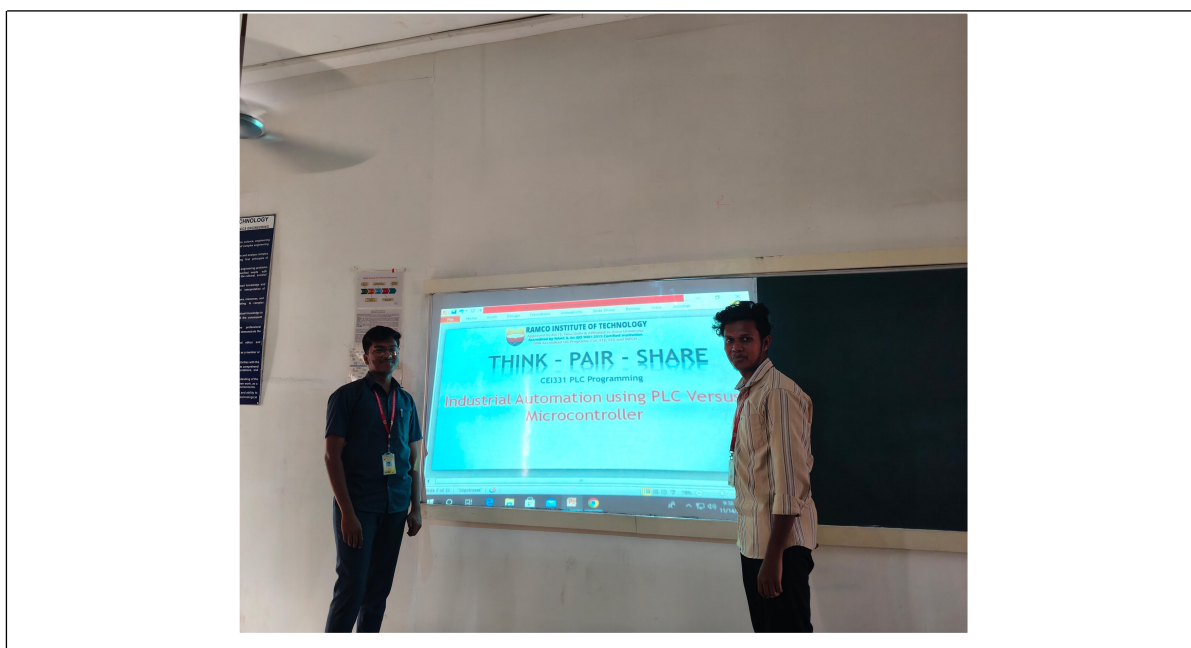
Benefit of the Practice

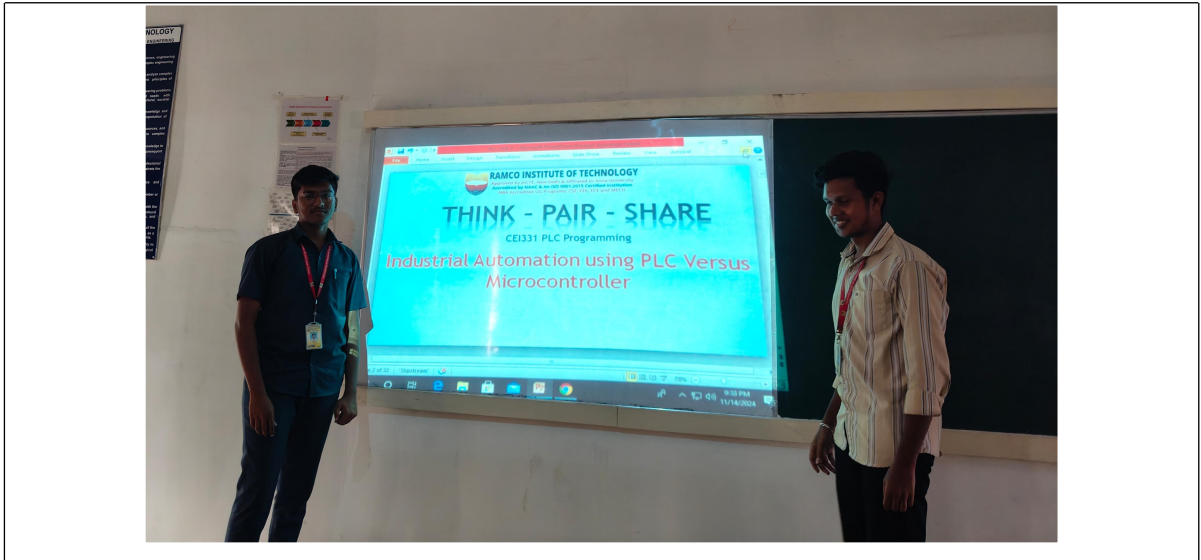
This activity fosters deeper understanding through peer collaboration, enhances communication skills, and encourages students to think critically about complex concepts. It allows for diverse perspectives and helps clarify misconceptions. Additionally, it promotes active engagement, making learning more interactive and enjoyable.

Challenges Faced in Implementation

Some students struggled to articulate their thoughts clearly during the initial thinking phase. Ensuring all pairs actively participated required monitoring and occasional prompting. Time constraints limited the depth of discussion, preventing some students from fully exploring the topic.

(v) Implementation Photograph:





References:

- ❖ A. Shunmugalatha, B. Ashok Kumar, S. Senthilrani, T. Chandrasekar, J. Rajeswari, “Smart Grid”, March 2021
- ❖ Joeri Van Mierlo, “Plug-in Hybrid Electric Vehicle (PHEV)”, Mdpi AG, 2019

Signature of Faculty Member

HOD